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Class: BSCS(6)
Section: 'A'

~~Outlook Temp Hot Windy Rain~~

	A Outlook	B Temp	C Humidity ~~Hot~~	D Windy	Rain
1	Sunny	Hot	High	Weak	No
2	Sunny	Mild	High	strong	No
3	Cloudy	Hot	High	weak	Yes
4	Rainy	Mild	High	weak	Yes
5	Rainy	cool	Normal	weak	Yes
6	Rainy	cool	Normal	Strong	No
7	Cloudy	Mild	Normal	Strong	Yes
8	Sunny	cool	Normal	weak	No

Total rows = 8
Yes = 4
No = 4

Prime attribute: Rain

Entropy (Rain):

$$-\frac{4}{8}\log_2\frac{4}{8} - \frac{4}{8}\log_2\frac{4}{8}$$

$$= -0.5\left(\frac{\log(0.5)}{\log(2)}\right) - 0.5\left(\frac{\log(0.5)}{\log(2)}\right)$$

$$= 0.5 + 0.5$$

$$= 1$$

A. Outlook

(1) Entropy:

(a) Sunny:

$$-\frac{0}{3}\log_2\frac{0}{3} - \frac{3}{3}\log_2\frac{3}{3}$$

$$-0\left(\frac{\log(0)}{\log(2)}\right)$$

$$0 - 1\left(\frac{\log(1)}{\log(2)}\right)$$

$$= 0$$

(b) Cloudy:

$$-\frac{2}{2}\log_2\frac{2}{2} - \frac{0}{2}\log_2\frac{0}{2}$$

$$-1\left(\frac{\log(1)}{\log(2)}\right) - 0\left(\frac{\log(0)}{\log(2)}\right)$$

$$= 0 - 0$$

$$= 0$$

(3) Rainy:

$$= -\frac{2}{3}\log_2\frac{2}{3} - \frac{1}{3}\log_2\frac{1}{3}$$

$$= -0.66\left(\frac{\log(0.66)}{\log(2)}\right) - 0.33\left(\frac{\log(0.33)}{\log(2)}\right)$$

$$= 0.399 + 0.527$$

$$= 0.926$$

(2) Weighted Entropy:

$$= \frac{3}{8}\times 0 + \frac{2}{8}\times 0 + \frac{3}{8}\times 0.926$$

$$= 0.375 \times 0.926$$

$$= 0.34725$$

(3) Info gain:

$$= 1 - 0.3475$$

$$IG = 0.625$$

B. Temperature

(1) Entropy:

(a) Hot:

$$-\frac{1}{2}\log_2\left(\frac{1}{2}\right) - \frac{1}{2}\log_2\frac{1}{2}$$

$$-0.5\left(\frac{\log(0.5)}{\log(2)}\right) - 0.5\left(\frac{\log(0.5)}{\log(2)}\right)$$

$= 0.5 + 0.5$

$= 1$

(b) Mild:

$= -\frac{2}{3}\log_2\frac{2}{3} - \frac{1}{3}\log_2\frac{1}{3}$

$= -0.66\left(\frac{\log(0.66)}{\log(2)}\right) - 0.33\left(\frac{\log(0.33)}{\log(2)}\right)$

$= 0.387 + 0.527$

$= 0.907$

(c) cool:

$-\frac{1}{3}\log_2\frac{1}{3} - \frac{2}{3}\log_2\frac{2}{3}$

~~0.332~~

$0.532 + 0.387$

0.1449

(d) Humidity:

(2) Weighted Entropy:

$\frac{2}{8} \times 1 + \frac{3}{8} \times 0.907 + \frac{3}{8} \times 0.1449$

$0.25 + 0.34 + 0.054$

$= 0.644$

(3) Info Gain $= 1 - 0.644$

$\boxed{IG = 0.356}$

Windy:

(1) Entropy:

(a) Weak: (Total = 5)

$$= -\frac{3}{5}\log_2\frac{3}{5} - \frac{2}{5}\log_2\frac{2}{5}$$

$$= 0.442 + 0.528$$

$$= 0.97$$

(b) Strong:

$$= -\frac{1}{3}\log_2\frac{1}{3} - \frac{2}{3}\log_2\frac{2}{3}$$

$$= 0.523 + 0.399$$

$$= 0.922$$

(2) Weighted Entropy:

$$\frac{5}{8} \times (0.97) + \frac{3}{8} \times (0.922)$$

$$0.606 + 0.345$$

$$= 0.951$$

(3) $IG = 1 - 0.951$

$$\boxed{IG = 0.045}$$

(c) Humidity

(1) Entropy:

(a) High:

$$= -\frac{2}{4} \times \log_2\left(\frac{2}{4}\right) - \frac{2}{4} \log_2 \frac{2}{4}$$

$$= 0.5 + 0.5$$

$$= 1$$

(b) Normal

$$= -\frac{2}{4} \log_2 \frac{2}{4} - \frac{2}{4} \log_2 \frac{2}{4}$$

$$= 0.5 + 0.5$$

$$= 1$$

(2) Weighted Entropy:

$$\frac{4}{8} \times (1) + \frac{4}{8} \times 1$$

$$0.5 + 0.5$$

$$= 1$$

Info Gain $= 1 - 1$

$= 0$

(7)

Attributes	weighted E	IG
Outlook	0.347	0.625
Temp	0.644	0.356
Humidity	0.951	0.045
Windy	1	0

Highest IG is 0.625
so we will check outlook

